FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – X
PAPER –1
TEST DATE: 15-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A (01 –03, 18 – 20, 35 – 37): *This section contains **NINE (9)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).*

Section – A (04 – 07, 21 – 24, 38 – 41): *This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

Section – A (08 – 11, 25 – 28, 42 – 45): *This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.*

Section – B (12 – 17, 29 – 34, 46 – 51): *This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

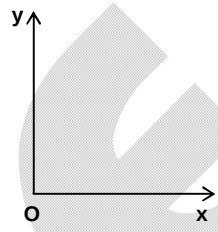
Physics

PART – I

SECTION – A
(One or More than one correct type)

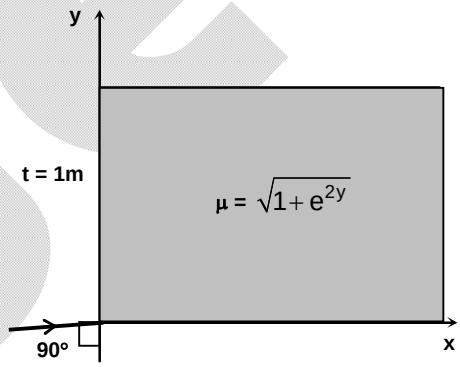
*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. Two coherent sources A and B emitting light of wavelength λ are placed at positions $(-D, 0)$ and $(-D, 3\lambda)$ respectively $(D >>> \lambda)$ as shown in the figure, then choose the correct option(s).



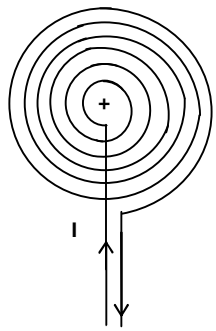
(A) The number of minima on the y-axis is 6 (+ve as well as –ve y-axis)
(B) The number of minima is more than the number of maxima on the y-axis (+ve as well as –ve y-axis)
(C) The number of maxima on the x-axis is 3
(D) The number of maxima on the x-axis is more than the number of minima on the x-axis

2. A ray of light travelling in air is incident at grazing angle on a long rectangular slab of thickness t = 1m at the origin. The refractive index of the transparent slab varies as $\mu = \sqrt{1+e^{2y}}$. Then, choose the correct option(s).



(A) The equation of trajectory of light in the slab is $x = (1 - e^{-y})$.
(B) The equation of trajectory of light in the slab is $x = (1 - e^{-2y})$.
(C) The co-ordinates of the point 'P' at which the light emerges from the slab are $\left(\frac{e+1}{e}, 1\right)$
(D) The co-ordinates of the point 'P' at which the light emerges from the slab are $\left(\frac{e-1}{e}, 1\right)$

3. Thin insulated wire forms a plane spiral consisting of a large number N of closely packed in turns through which a DC current I is flowing. The radii of internal and external loops are a and b respectively. Choose the correct option(s).



(A) The magnetic field at the centre of the spiral is $\frac{\mu_0 IN}{2(b-a)} \ell n \frac{b}{a}$
(B) The magnetic field at the centre of the spiral is $\frac{\mu_0 IN}{(b-a)} \ell n \frac{b}{a}$
(C) Magnetic moment of the spiral is $\frac{I\pi N}{2}(b^2 + a^2 + ab)$
(D) Magnetic moment of the spiral is $\frac{I\pi N}{3}(b^2 + a^2 + ab)$

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

4. A sinusoidal input voltage source v = (2 + 3sinωt + 4 cos ωt) volt is connected across a load, then choose the correct option.

(A) RMS value is $\left(\sqrt{\frac{33}{2}}\right)$ V (B) RMS value is $\left(\frac{2\sqrt{2}+7}{\sqrt{2}}\right)$ V

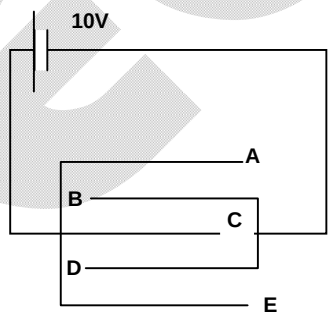
(C) Average value is zero (D) Average value is 5V

5. The battery connected across a coolidge tube operates at a power level of 1W when the no. of electrons hitting the target in 1 second is 6.25×10^{13}. Find the minimum wavelength in the resulting X-rays spectrum:

(A) 0.06 Å (B) 0.12 Å

(C) 0.48 Å (D) 0.24 Å

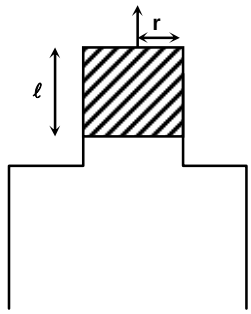
6. Five identical capacitor plates are arranged such that they make four capacitors each of 2μF. The plates are connected to a source of emf 10 V. The charge on plate C is



(A) 20 μC (B) 40 μC

(C) 60 μC (D) 80 μC

7. A cylindrical massless cork of length $\ell = 10$cm and radius r = 3cm is slowly extracted from the neck of a bottom opened bottle. If the normal pressure between the bottle and un-extracted part of the cork at any instant is constant and equal to $P = \frac{10^5}{\pi}$ N/m^2. The work done in extracting it completely is



(Take the coefficient of friction between the cork and bottle is $\mu = 0.3$)

(A) 5 J (B) 9 J

(C) 12 J (D) 15 J

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

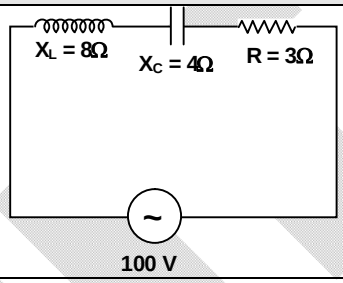
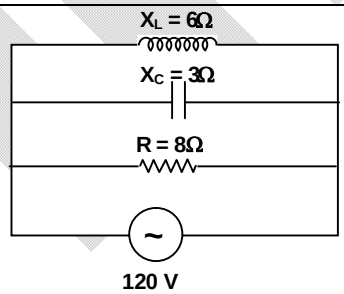
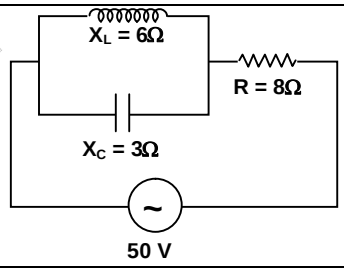
8. Components along and perpendicular to a position vector are known as radial and transverse components respectively. A particle is projected horizontally from the top a tower. Assume acceleration due to gravity to be uniform and the origin of the coordinate system at the point of projection.

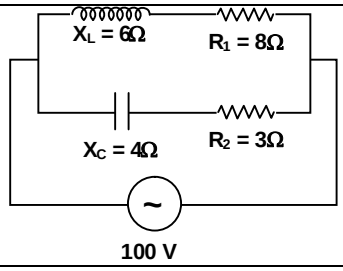
List –I		List –II	
(P)	Radial component of velocity	(1)	Always increases.
(Q)	Transverse component of velocity	(2)	Always decreases.
(R)	Radial component of acceleration	(3)	First increases then decreases.
(S)	Transverse component of acceleration	(4)	First decreases then increases.
		(5)	Remains Constant.

The correct option is:

(A) (P) → (1) (Q) → (1) (R) → (1) (S) → (2)
(B) (P) → (2) (Q) → (1) (R) → (1) (S) → (4)
(C) (P) → (1) (Q) → (2) (R) → (1) (S) → (4)
(D) (P) → (1) (Q) → (1) (R) → (3) (S) → (1)

9. List –I contains different AC circuits with rms voltage of ac sources as shown. List –II contains the corresponding rms currents supplied by the ac voltage sources. Then match the options of List-I correctly with the options of List-II.

List –I		List –II	
(P)		(1)	$I_{rms} = 25$ A
(Q)		(2)	$I_{rms} = 5$ A
(R)		(3)	$I_{rms} = 10\sqrt{5}$ A

(S)		(4)	I_{rms} = 20 A
		(5)	I_{rms} = 30 A

The correct option is:

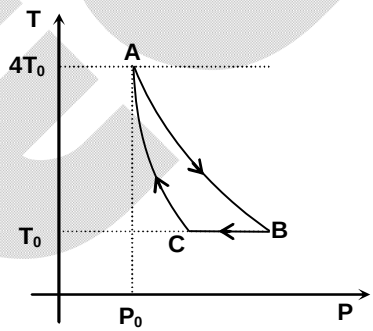
(A) (P) → (4) (Q) → (1) (R) → (2) (S) → (3)

(B) (P) → (4) (Q) → (3) (R) → (1) (S) → (2)

(C) (P) → (3) (Q) → (2) (R) → (5) (S) → (1)

(D) (P) → (1) (Q) → (4) (R) → (2) (S) → (5)

10. One mole of an ideal diatomic gas undergoes a cyclic process as shown in Temperature-Pressure graph. Temperature and pressure of gas at 'A' is $4T_0$ and P_0 respectively and temperature in process BC is T_0. The process AB is given by TP^2 = constant and the process CA is given by TP^4 = constant.



List –I describe the name of thermodynamics process involve in the cyclic process and List-II gives work done by gas in the process, charge in internal energy of the gas in the process and heat given to the gas in the process.

List –I		List –II	
(P)	In the thermodynamics process A → B	(1)	Work done by gas and heat given to gas is negative
(Q)	In the thermodynamics process B → C	(2)	Work done by gas is negative but heat given to gas is positive
(R)	In the thermodynamics process C → A	(3)	Work done by gas is positive but heat given to gas negative
(S)	In the cyclic process	(4)	Work done by gas and heat given to gas is positive
		(5)	Work done by gas is positive and change in internal energy of gas is negative

The correct option is:

(A) (P) → (1) (Q) → (4) (R) → (4) (S) → (1)

(B) (P) → (3) (Q) → (4) (R) → (1) (S) → (1)

(C) (P) → (3) (Q) → (4) (R) → (4) (S) → (1)

(D) (P) → (1) (Q) → (4) (R) → (2) (S) → (1)

11. Match the statements in List-I with the values in List-II.

List –I		List –II	
(P)	A stationary source emits sound of frequency 660 Hz. The sound is reflected by a reflector moving away from the source with a velocity 33 m/s. The frequency (in Hz) of the sound reflected by the reflector is (speed of sound in air is 330 m/s)	(1)	600
(Q)	A bus is moving towards a hill with speed 16.5 m/s. The driver of the bus sounds a horn of frequency of 627 Hz. The frequency (in Hz) of the echo heard by a passenger in the bus will be (speed of sound in air is 330 m/s)	(2)	500
(R)	A fighter plane is travelling in a horizontal direction at a certain height from the ground with a speed equal to half the speed of sound in air. The frequency of the sound produced by the plane is 450 Hz. What is the frequency (in Hz) of the sound heard by a stationary observer on the ground when the plane is directly above the observer.	(3)	540
(S)	A star emits a electromagnetic wave of frequency $500\sqrt{3}$ GHz. It moves away from a observer with a relative speed equal to half of the speed of light in vacuum. The frequency of the wave (in GHz) observed by the observer is	(4)	620
		(5)	693

The correct option is:

(A) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (4) (S) $\rightarrow$ (1)

(B) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (5) (R) $\rightarrow$ (1) (S) $\rightarrow$ (1)

(C) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (5) (R) $\rightarrow$ (1) (S) $\rightarrow$ (2)

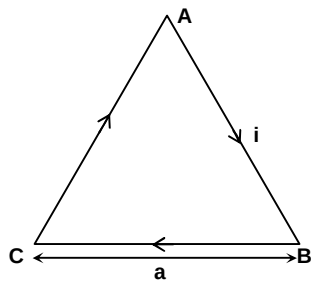
(D) (P) $\rightarrow$ (1) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (2) (S) $\rightarrow$ (1)

SECTION – B

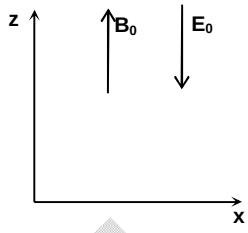
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

12. A wire bent in the form of an equilateral triangle ABC of side a carries a current I. The magnetic field at a point having equal distance a from A, B, and is $\frac{\mu_0 i}{2\sqrt{k\pi a}}$. Find k.



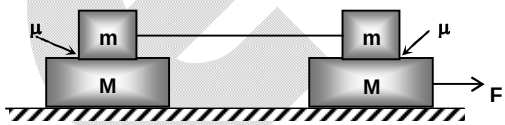
13. In a certain region, uniform electric field $E = -E_0\hat{k}$ and magnetic field $B = B_0\hat{k}$ are present. At time t = 0, a particle of mass m and charge q is given a velocity $v = v_0\hat{j} + v_0\hat{k}$. Find the time (in sec) when speed of particle gets minimum. (Given $mv_0 = 4qE_0$)



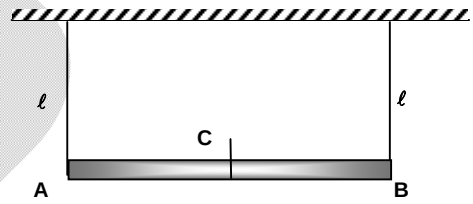
14. Two identical thin converging lenses are kept in contact at a distance 12.5 cm from an object. If real image produced is 4 times enlarged then what is the optical power of each lens in dioptres.

15. Consider the following nuclear reaction
$X^{200} \rightarrow A^{110} + B^{90} + \text{Energy}$
If the binding energy per nucleon for X, A and B are 7.4 MeV, 8.2 MeV and 8.2 MeV respectively, the energy released is 40K in MeV. Find the value of K.

16. In the figure, ground is smooth, coefficient of friction between blocks m and M is 'μ'. Find the maximum value of F, such that there is no slipping between blocks m and M in newton. (m = 10 kg, M = 40 kg, μ = 0.3)



17. A uniform rod AB of mass m is suspended by two identical string of length ℓ as shown in the figure. If this rod is turned by a small angle in horizontal plane about, the vertical axis passing through its centre C. In this process the strings are deviated by a small angle θ_0 from vertical. Then the rod is released to start performing angular oscillations. The period of oscillation is $2\pi\sqrt{\frac{\ell}{\lambda g}}$. Find the value of λ.



Chemistry

PART – II

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

18. Which of the following statement is/are **incorrect?**

(A) For solid and liquid $\left(\frac{\partial H}{\partial T}\right)_P > \left(\frac{\partial U}{\partial T}\right)_{Vs}$.

(B) The gas following the equation $P(V-b)=RT$ cannot be liquefied.

(C) The total energy of the ideal gas molecule is the kinetic energy.

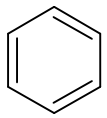
(D) Below critical temperature, Z > 1.

19. Which of the following compounds would undergo aldol condensation?

(A) 3-methyl pentanal (B) 1-phenyl propanone

(C) phenyl acetaldehyde (D) 2,2-dimethyl butanal

20. The correct statement(s) is/are about product 'R'



$$\xrightarrow[\substack{2.\ m-CPBA \\ 3.\ NaOH, H_2O \\ 4.\ H^+}]{1.\ MeCOCl/AlCl_3} P + Q$$

$$P \xrightarrow{NaHCO_3} CO_2$$

$$Q \xrightarrow[CO_2/H^+]{NaOH} R(\text{Major})$$

(A) R gives white ppt. with bromine water.

(B) R is more acidic than its corresponding position isomer

(C) $R \xrightarrow[Py]{Ac_2O}$ Aspirin

(D) $R \xrightarrow[2.\ pH=8,\ PhN_2Cl]{1.\ NaOH+CaO,\ \Delta}$ Colour compound

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

21. The green colour produced in the borax bead test of a chromium (III) salt is due to

(A) $Cr(BO_2)_3$ (B) $Cr_2(B_4O_7)_3$

(C) Cr_2O_3 (D) CrB

22. A dilute aqueous solution of Na_2SO_4 is electrolysed using Pt electrode. The products at the anode and cathode respectively are:

(A) O_2, H_2 (B) $S_2O_8^{2-}$, Na

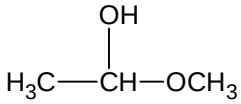
(C) O_2, Na (D) $S_2O_8^{2-}, H_2$

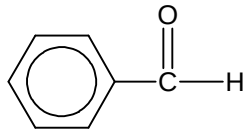
23. Magnitude of heat of neutralization for weak acid HF with strong base is

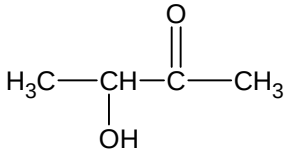
(A) 13.7 Kcal (B) > 13.7 Kcal

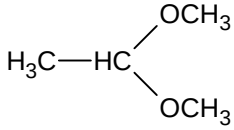
(C) < 13.7 Kcal (D) None of these

24. Which of the following will not give positive Tollen's test?

(A) 

(B) 

(C) 

(D) 

SECTION – A
(Matching List Type)

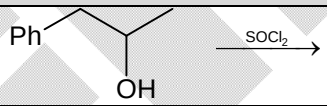
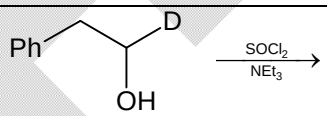
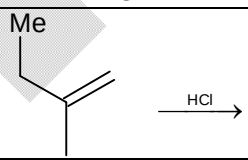
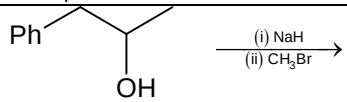
*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match List –I with List-II:

List – I		List – II	
(P)	Schimdt reaction	(1)	$RCOOH \xrightarrow[\Delta]{NaOH/CaO} RH$
(Q)	Curtius reaction	(2)	$R-CH_2COOH \xrightarrow[\Delta]{Red\ P/X_2} R-\underset{X}{CH}-COOH$
(R)	Decarboxylation	(3)	$RCOCl \xrightarrow[(ii)\ H_3O^+]{(i)\ NaN_3/\Delta} R-NH_2$
(S)	HVZ reaction	(4)	$RCOOH \xrightarrow[\Delta]{N_2H} R-NH_2$
		(5)	$CH_3CHO \xrightarrow[\Delta]{dil.\ OH^-} CH_3-CH=CH-CHO$

(A) P → 4; Q → 3; R → 1; S → 2
(B) P → 4; Q → 3; R → 5; S → 1
(C) P → 5; Q → 1; R → 2; S → 3
(D) P → 5; Q → 4; R → 1; S → 2

26. Match List –I with List-II:

List – I (Reaction)		List – II (Comment on major product and reaction intermediate)	
(P)		(1)	Optically active compound
(Q)		(2)	Inversion of configuration
(R)		(3)	Retention of configuration
(S)		(4)	Optically inactive compound
		(5)	Carbocation intermediate

(A) P → 1; Q → 3; R → 4; S → 2
(B) P → 3; Q → 1; R → 2; S → 4
(C) P → 3; Q → 2; R → 4; S → 1
(D) P → 4; Q → 1; R → 5; S → 2

27. Match List –I with List-II:

List – I (Reaction)		List – II (Characteristics of product)	
(P)	$P_2O_3 + H_2O \longrightarrow$	(1)	One P – OH Two P – H One P = O
(Q)	Red P_4 + alkali $\longrightarrow$	(2)	Four P – OH Two P = O One P – P
(R)	White P_4 + alkali $\longrightarrow$	(3)	Two P – OH Two P – H Two P = O
(S)	$PCl_3 + H_3PO_3 \longrightarrow$	(4)	Two P – OH One P – H One P = O
		(5)	Four P – OH Two P = O Two P – O – P

(A) P → 2; Q → 5; R → 1; S → 3
(B) P → 5; Q → 2; R → 1; S → 3
(C) P → 4; Q → 1; R → 3; S → 2
(D) P → 4; Q → 2; R → 1; S → 3

28. Match List –I with List-II:

List – I (Complex ion)		List – II (Number of unpaired electrons)	
(P)	$\left[Co(NH_3)_6\right]^{2+}$	(1)	Zero
(Q)	$\left[Ni(NH_3)_6\right]^{2+}$	(2)	One
(R)	$\left[Fe(H_2O)_6\right]^{3+}$	(3)	Five
(S)	$\left[Fe(CN)_5(NO)\right]^{2-}$	(4)	Two
		(5)	Three

(A) P → 5; Q → 4; R → 3; S → 1
(B) P → 5; Q → 2; R → 4; S → 1
(C) P → 1; Q → 5; R → 3; S → 2
(D) P → 2; Q → 4; R → 5; S → 3

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

29. Number of 90° bond angles present in the molecule XeO_6^{4-} ion is X and number of 90° bond angles present in PCl_3F_2 is Y and 90° bond angles in ClF_3 is Z. Then X – Y – Z is

30. P and Q are ideal gases that do not react each other. The mass of 1 mole of P is four times that of Q. At standard temperature and pressure, how many of the following are correct based on above data.
(i) Average kinetic energy of a molecule of P is equal to that of molecules of Q.
(ii) The mass of 1 dm^3 of P is four times that of 1 dm^3 of Q.
(iii) On mixing 1 dm^3 of P with 1 dm^3 of Q, the partial pressure of each gas in the mixture will be 0.5 atm.
(iv) The diffusion rate of P is four times that of Q.

31. $XeF_6 \xrightarrow[\text{hydrolysis}]{\text{partial}} A$ $\quad XeF_6 \xrightarrow[\text{hydrolysis}]{\text{complete}} B$

The sum of lone pairs on he central atoms in Xenon compound A and B are

32. Among B_2H_6, $B_3N_3H_6$, N_2O, N_2O_4, $H_2S_2O_3$ and $H_2S_2O_6$, the number of molecules containing covalent bond between two atoms of the same kind is

33. $CaC_2 \xrightarrow{H_2O}$ A $\xrightarrow[\text{heating}]{\text{Red hot/Cu}}$ B $\xrightarrow[AlCl_3]{CH_3-\overset{\overset{O}{\|}}{C}-Cl}$ C $\xrightarrow{NO_2^+BF_4^-}$ D $\xrightarrow{Sn/HCl}$ E $\xrightarrow[\text{(iii) } H_3PO_2]{\text{(i) NaOCl (ii) } NaNO_2/HCl}$ F

Organic product

Sum of degree of unsaturation D and F is

34. The vapour pressure of two pure liquid A and B, that form an ideal solution is 100 mm Hg and 400 mm Hg respectively at temperature T. This liquid solution of A and B is composed of 1 mol of A and 1 mol of B. What will be the final vapour pressure when 1 mole of the mixture has been vapourised?

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

35. Each of 2024 boxes in a line contains one red marble, and for $1 \le k \le 2024$, the box in the kth position also contains k white marbles. A child begins at the first box and successively draws a single marble at random from each box in order. He stops when he first draws a red marble. Let P(n) be the probability that he stops after drawing exactly n marbles. The possible value(s) of n for which $P(n) < \frac{1}{2024}$ is

(A) 44 (B) 45
(C) 46 (D) 47

36. y = f(x) is a parabola having directrix with slope 0. If $f^2(0) + f^2(1) + f^2(2) + 45 = 10f(0) + 8f(1) + 4f(2)$, then

(A) y = f(x) is symmetric about $x = -\frac{1}{2}$ (B) $f(x) < 0 \ \forall \ x \in R$

(C) $\ln|f(x)|$ is one-one function (D) range of f(x) is $\left(-\infty, \frac{41}{8}\right]$

37. Let we have to construct an $n \times n$ square array from the numbers –1, 0, 1 such that all the sums, namely rows, columns and both diagonal sums are distinct, then which of the options are incorrect values of n

(A) 1 (B) 2
(C) 3 (D) 0

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

38. If $^{2024}C_{1012}$ can be prime factorized as $2^a\ 3^b\ 5^c\ 7^d$..... where a, b, c, d are non-negative integers, then a + b + c + d is equal to

(A) 7 (B) 6
(C) 11 (D) 9

39. Given that $x_1 = 211$, $x_2 = 375$, $x_3 = 420$, $x_4 = 523$, $x_5 = 267$ and $x_n = x_{n-1} - x_{n-2} + x_{n-3} - x_{n-4}$ (where $n \ge 5$), then the value of $|x_{2025} + x_{2026} + x_{2027}|$ is

(A) 319 (B) 320
(C) 321 (D) 322

40. Let z_1, z_2, z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = R$ and $z_2 \ne z_3$, then minimum value of $|az_2 + (1 - a)z_3 - z_1|$ is (where $a \in R$)

(A) $\frac{1}{2R}|z_2 - z_3| \cdot |z_1 - z_3|$ (B) $\frac{1}{2R}|z_1 - z_2| \cdot |z_1 - z_3|$

(C) $\frac{1}{4R}|z_2 - z_3| \cdot |z_1 - z_3|$ (D) $\frac{1}{4R}|z_1 - z_2| \cdot |z_1 - z_3|$

41. For positive numbers a, b, c, d, e if a + b + c + d + e = 1, then minimum value of $\frac{1-a}{1+a}+\frac{1-b}{1+b}+\frac{1-c}{1+c}+\frac{1-d}{1+d}+\frac{1-e}{1+e}$ is

(A) $\frac{20}{3}$ (B) $\frac{5}{3}$

(C) $\frac{6}{4}$ (D) $\frac{10}{3}$

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

42. Match the following List-I with List-II

List – I		List - II	
(P)	If $\int_0^1 \left(f^6(x)-8xf^3(x)\right)dx+\frac{16}{3}=0$, then $f^3(1)$ is	(1)	0
(Q)	If $g(x)=\int_0^x\left(t^4\int_1^x f(u)du\right)dt$, then g′(1) + g″(1) is $k_1+\frac{f'(1)}{k_2}$, then k_1-k_2 (given f(1) = 5)	(2)	1
(R)	The value of $\int_0^1 \frac{t^3-1}{\ln t}dt$ is	(3)	3
(S)	Let $I_1=\int_0^1 \frac{e^t}{1+t}dt$ and $I_2=\int_0^1 \frac{t^2}{e^{t^3}(2-t^3)}dt$, then $\frac{I_1}{eI_2}$ is	(4)	4
		(5)	6

The correct option is:
(A) (P) → (2); (Q) → (4); (R) → (3); (S) → (1)
(B) (P) → (2); (Q) → (3); (R) → (2); (S) → (1)
(C) (P) → (4); (Q) → (5); (R) → (2); (S) → (3)
(D) (P) → (4); (Q) → (5); (R) → (1); (S) → (3)

43. Match the following List-I with List-II

List – I		List – II	
(P)	If $P=(1,7,\sqrt{2})$ be a point and the line L, $2\sqrt{2}(x-1)=y-2$, z = 0. If PQ is the distance from plane $\sqrt{2}x+y-z=1$ from point P measured along a line inclined at angle 45° with the line L and is minimum, then the value of PQ is	(1)	0
(Q)	If $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}+2\hat{k}$ and $\lvert\vec{c}\rvert=1$ such that $\left[\vec{a}\times\vec{b}\quad\vec{b}\times\vec{c}\quad\vec{c}\times\vec{a}\right]$ has a maximum value, then the value of $\left\lvert(\vec{a}\times\vec{b})\times\vec{c}\right\rvert^2$ is	(2)	3

(R)	If $\left[\vec{a}+2\vec{b}+3\vec{c} \quad \vec{b}+2\vec{c}+3\vec{a} \quad \vec{c}+2\vec{a}+3\vec{b}\right]=36$ where $\vec{a},\vec{b}$ and $\vec{c}$ are three vectors, then the value of $\begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c}\\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c}\end{vmatrix}$ is equal to	(3)	5
(S)	The plane $\left(a\hat{i}-\hat{j}+7\hat{k}\right)\cdot\vec{r}=1$, $\left(a\hat{i}+\hat{j}-2\hat{k}\right)\cdot\vec{r}=-1$ and $\left(2\hat{i}-\hat{j}+2\hat{k}\right)\cdot\vec{r}=5$ intersect the plane $\hat{k}\cdot\vec{r}=0$ along the lines DC, EC and ED respectively. If CD subtends right angle at the origin, then value of 10a – 3 is equal to	(4)	4
		(5)	1

The correct option is:
(A) (P) → (4); (Q) → (3); (R) → (2); (S) → (1)
(B) (P) → (2); (Q) → (1); (R) → (4); (S) → (5)
(C) (P) → (4); (Q) → (3); (R) → (2); (S) → (5)
(D) (P) → (2); (Q) → (3); (R) → (4); (S) → (3)

44. Match the following List-I with List-II

List – I		List – II	
(P)	Let S be the sample space of all 3×3 matrices with entries from the set {0, 1}. Let the event E_1 and E_2 be given by E_1 = {A ∈ S : Det A = 0} and E_2 = {A ∈ S : Sum of entries of A is 7}. If a matrix is chosen at random from S and $P\left(\frac{E_1}{E_2}\right)=\frac{p}{q}$ where HCF (p, q) = 1, then value of p + q is	(1)	1
(Q)	The value of $\begin{vmatrix}1 & a & a^2-bc\\ 1 & b & b^2-ca\\ 1 & c & c^2-ab\end{vmatrix}$ is equal to	(2)	2
(R)	If P, Q, R are the angles of triangle PQR, then the value of $\begin{vmatrix}-1 & \cos R & \cos Q\\ \cos R & -1 & \cos P\\ \cos Q & \cos P & -1\end{vmatrix}$ is equal to	(3)	3
(S)	A is the $n \times n$ matrix whose elements are all 1 and B is the $n \times n$ matrix whose diagonal elements are all n and the other elements are (n – r), then $4 + |(B - rI)(B - (n^2r - nr + r)I)|$ is	(4)	4
		(5)	0

The correct option is:
(A) (P) → (3); (Q) → (5); (R) → (2); (S) → (4)
(B) (P) → (4); (Q) → (3); (R) → (2); (S) → (1)
(C) (P) → (3); (Q) → (5); (R) → (1); (S) → (4)
(D) (P) → (4); (Q) → (3); (R) → (5); (S) → (1)

45. Match the following List-I with List-II

List – I		List – II	
(P)	On the ellipse $\frac{x^2}{64}+\frac{y^2}{9}=1$, tangents drawn at P_1, P_2, P_3 ..... P_n intersects the major axis T_1, T_2, T_3 ....., T_n respectively. If the value of $\sum_{i=1}^{n}\frac{\left(\text{Area}\left(\Delta P_iT_iS\right)\Delta\left(P_iT_iS'\right)\right)}{\left(P_iT_i\right)^2}=18$, then n is equal to (where S and S′ represents the two focus of ellipse)	(1)	8
(Q)	If one of three normals drawn from point P(8, – 2) to parabola $y^2 = 4ax(0 < a < 3)$ passes through (3a, 0). If length of latus rectum of parabola is k, then k – 4 is	(2)	5
(R)	Consider points P(7, 1) and O = (0, 0) if S is a point on the line y = x and T is a point on the x-axis so that P is on the line segment ST, then the minimum possible area of ΔOST is	(3)	4
(S)	The line x + 2y = a intersects the circle $x^2 + y^2 = 4$ at two distinct points A and B. Another line 12x – 6y – 41 = 0 intersects the circle $x^2 + y^2 – 4x – 2y + 1 = 0$ at two another distinct points C and D. the value of |a| for which A, B, C and D are concyclic	(4)	12
		(5)	2

The correct option is:
(A) (P) → (1); (Q) → (3); (R) → (4); (S) → (4)
(B) (P) → (2); (Q) → (3); (R) → (5); (S) → (1)
(C) (P) → (2); (Q) → (3); (R) → (1); (S) → (4)
(D) (P) → (1); (Q) → (3); (R) → (4); (S) → (5)

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

46. Let α_n be the coefficient of x^2 in $\left(2-\sqrt{x}\right)^n$, then find $3\lim_{n\to\infty}\left[\frac{2}{\alpha_4}+\frac{2^2}{\alpha_5}+\frac{2^3}{\alpha_6}+\frac{2^4}{\alpha_7}+.....\infty\right]$

47. A quadratic equation with roots α and β where $\alpha\beta = 4$, satisfies the equation $\frac{\alpha}{\alpha-1}+\frac{\beta}{\beta-1}=\frac{a^2-7}{a^2-4}$. Number of integral values of a for which $\alpha, \beta \in (1, 4)$ is equal to

48. The terms of in arithmetic sequence add to 465. The first term of sequence is increased by 1, the second term is increased by 3 and so on, in general the kth term is increased by the kth odd positive integers. The terms of new sequence add to 690. If t_1 and t_n denotes first and last term of the original sequence respectively, then $\frac{\left(t_1+t_n\right)}{31}$ is

49. If three different polynomials $x^2 + ax + b$, $x^2 + x + ab$ and $ax^2 + x + b$ have exactly one common zero, where a, b are non-zero real numbers, then the values of a + 2b is

50. The equation $z^4 + (5z - 1)^4 = 0$ has two pairs of complex roots α_1, β_1 and α_2, β_2. Each pair α_i, β_i for i = {1, 2} are complex conjugates. Then value of $\frac{1}{13}\sum_{i=1}^{2}\frac{1}{\alpha_i\beta_i}$

51. $f(x)=\begin{cases}0 & ; \quad x<\frac{1}{2}\\ 1 & ; \quad x\geq\frac{1}{2}\end{cases}$. If the value of $\int_0^1 f(\sin\pi x)\,dx=\frac{p}{q}$ (where p and q) are relative prime, then the value of p + q is